

History

Four 12-gauge shotgun shells are shown side-by-side. From left to right: a standard slug shell with a conical slug, a standard shot shell with a cylindrical body and a visible shot column, a sabot shell with a long cylindrical body and a conical slug inside, and a sabot slug with a long cylindrical body and a conical slug inside.

.40 S&W	
	
.40 S&W FMJ flat-point cartridge	
Type	Pistol
Place of origin	United States
Production history	
Designer	Bob Klunk
Designed	January 17, 1990
Produced	1990–present
Specifications	
Parent case	10mm Auto
Case type	Rimless, straight
Bullet diameter	.400 in (10.2 mm)
Land diameter	.390 in (9.9 mm)
Neck diameter	.423 in (10.7 mm)
Base diameter	.424 in (10.8 mm)
Rim diameter	.424 in (10.8 mm)
Rim thickness	.055 in (1.4 mm)
Case length	.850 in (21.6 mm)
Overall length	1.135 in (28.8 mm)
Case capacity	19.3 gr H ₂ O (1.25 cm ³)
Rifling twist	1 in 16 in. (406 mm)
Primer type	Small pistol
Maximum pressure	35,000 psi (240 MPa)
Ballistic performance	

During tests of the 9×19 mm and .45 ACP ammunition, the Unit Chief of the FBI Firearms Training Unit, John Hall, decided to include tests of the 10mm Auto cartridge, supplying his own Colt Delta Elite 10 mm semi-automatic, and personally hand-loaded ammunition. The FBI's tests revealed that a 170–180 gr (11.0–11.7 g) JHP 10 mm bullet, propelled between 900–1,000 ft/s (270–300 m/s), achieved desired terminal performance without the heavy recoil associated with conventional 10 mm ammunition (1,300–1,400 ft/s (400–430 m/s)). The FBI contacted Smith & Wesson and requested it to design a handgun to FBI specifications, based on the existing large-frame Smith & Wesson Model 4506 .45 ACP handgun, that would reliably function with the FBI's reduced-velocity 10 mm ammunition. During this collaboration with the FBI, S&W realized that downsizing the 10 mm full power to meet the FBI's medium velocity specification meant less powder and more airspace in the case. They found that by removing the airspace they could shorten the 10 mm case enough to fit within their medium-frame 9 mm handguns and load it with a 180 gr (11.7 g) JHP bullet to produce ballistic performance identical to the FBI's reduced-velocity 10 mm cartridge. S&W then teamed with Winchester to produce a new cartridge, the .40 S&W. It uses a small pistol primer whereas the 10 mm cartridge uses a large pistol primer.

Bullet mass/type	Velocity	Energy
10.69 g (165 gr)	1,130 ft/s	468 ft·lbf
Federal FMJ	(340 m/s)	(635 J)
7.45 g (115 gr) Cor-Bon	1,400 ft/s	500 ft·lbf
Glaser	(430 m/s)	(680 J)
10.04 g (155 gr)	1,160 ft/s	463 ft·lbf
Federal HST	(350 m/s)	(628 J)
8.74 g (135 gr)	1,400 ft/s	588 ft·lbf
Underwood JHP	(430 m/s)	(797 J)
12.95 g (200 gr)	1,050 ft/s	490 ft·lbf
Doubletap FMJ FP	(320 m/s)	(660 J)
<i>Test barrel length: 4 inches (100 mm)</i> <i>Source(s): ^[1]^[2]</i>		

The .40 S&W cartridge debuted January 17, 1990, along with the new Smith & Wesson Model 4006 pistol, although it was several months before the pistols were available for purchase. Austrian manufacturer Glock Ges.m.b.H. were commercially available ahead of Smith & Wesson in 1990, with pistols chambered in .40 S&W (the Glock 22 and Glock 23) which were announced a week before the 4006.^[6] Glock's rapid introduction was aided by its engineering of a pistol chambered in 10 mm Auto, the Glock 20, only a short time earlier. Since the .40 S&W uses the same bore diameter and case head as the 10 mm Auto, it was merely a matter of adapting the 10 mm design to the shorter 9×19mm Parabellum frames. The new guns and ammunition were an immediate success,^[7]^[8] and pistols in the new caliber were adopted by several law enforcement agencies around the nation, including the FBI, which adopted the Glock pistol in .40 S&W in May 1997.

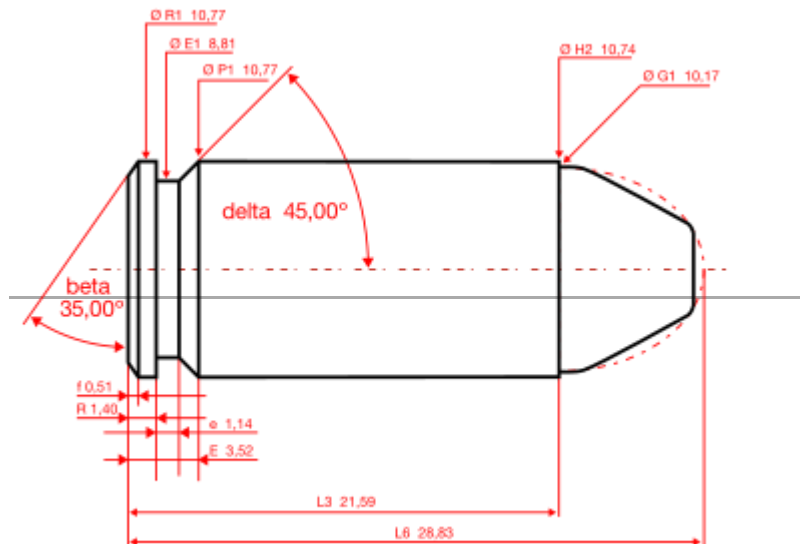
The popularity of the .40 S&W accelerated with the passage of the now-expired Federal Assault Weapons Ban of 1994 which prohibited sales of pistol or rifle magazines that could hold more than ten cartridges, regardless of caliber. Several U.S. states, and a number of local governments, also banned or regulated so-called "high capacity" magazines. As a result, many new firearm buyers limited to purchasing pistols with a maximum magazine capacity of 10 rounds chose pistols in the .40 S&W chambering instead of smaller-diameter cartridges such as the 9x19 mm (9 mm Luger or 9 mm Parabellum).

The .40 S&W case length and overall cartridge length are shortened, but other dimensions except for case web and wall thickness remain identical to the 10 mm Auto. Both cartridges headspace on the mouth of the case. Thus in a semi-auto, they are not interchangeable. Fired from a 10 mm semi-auto, the .40 Smith & Wesson cartridge will headspace on the extractor and the bullet will jump a 0.142 inches (3.6 mm) freebore just like a .38 Special fired from a .357 Magnum revolver. If the cartridge is not held by the extractor, the chances for a ruptured primer are great.^[9] Smith & Wesson does make a double-action revolver (the Model

610) that can fire either cartridge via use of moon clips. A single-action revolver in the .38–40 chambering can also fire .40 or 10 mm rounds provided it is equipped with a correctly sized cylinder. Some .40 caliber handguns can be converted to 9 mm with a special purpose-made barrel, magazine change, and other parts.

Cartridge dimensions

The .40 S&W has 1.25 ml (19.3 grains H₂O) cartridge case capacity.



The common rifling twist rate for this cartridge is 406 millimetres (16.0 in), 6 grooves, $\varnothing$ lands = 9.91 ;mm, $\varnothing$ grooves = 10.17 mm, land width = 3.05 mm and the primer type is small pistol.^[4] According to the official C.I.P. guidelines, the .40 S&W case can handle up to 225 megapascals (32,600 psi) piezo pressure. In C.I.P.-regulated countries, every pistol/cartridge combo has to be proofed at 130% of this maximum C.I.P. pressure to certify for sale to consumers. The SAAMI pressure limit for the .40 S&W is set at 241.32 megapascals (35,001 psi) piezo pressure.^[10]

Performance



An expanded hollow point bullet (Winchester SXT "Black Talon", left) and an unfired hollow point .40 S&W cartridge (Speer Gold Dot, right)

.40 S&W cartridge has been popular with law enforcement agencies in the United States, Canada, Australia, and Brazil. While possessing nearly identical accuracy,^[11] drift and drop as the 9 mm Parabellum, it also has an energy advantage^[12] over the 9 mm Parabellum^[13] and .45 ACP,^[14] and with a more manageable recoil than the 10 mm Auto cartridge.^[7] Marshall & Sanow (and other hydrostatic shock proponents) contend that with good jacketed hollow point bullets, the more energetic loads for the .40 S&W can also create hydrostatic shock in human-sized living targets.^{[15][16]}

Based on the ideal terminal ballistic performance in ordnance gelatin during lab testing in the late 1980s and early 1990s, the .40 S&W earned status as "the ideal cartridge for personal defense and law enforcement".^{[8][17]} Ballistically the .40 S&W is almost identical to the .38-40 Winchester introduced in 1874, as they share the same bullet diameter and bullet weight, and have similar muzzle velocities.^[18] The energy of the .40 S&W exceeds standard-pressure .45 ACP loadings, generating between 350 foot-pounds (470 J) and 500 foot-pounds (680 J) of energy, depending

on bullet weight. Both the .40 S&W and the 9 mm Parabellum operate at a 35,000 pounds per square inch (240 MPa) SAAMI maximum, compared to a 21,000 pounds per square inch (140 MPa) maximum for .45 ACP.^[19]

.40 S&W pistols with standard (not extended) double-stack magazines can hold as many as 16 cartridges. While not displacing the 9 mm Parabellum, the .40 S&W is commonly used in law enforcement applications in keeping with its origin with the FBI. Select U.S. special operations units have available the .40 S&W and .45 ACP for their pistols. The United States Coast Guard, having dual duties as maritime law enforcement and military deployments, has adopted the SIG Sauer P229R DAK in .40 S&W as their standard sidearm. However, as of 2020, the United States Coast Guard has adopted the Glock 19 GEN5 MOS as its new sidearm. Numerous federal and local law enforcement agencies are currently in the process of transitioning from .40 S&W to 9 mm, most notably the FBI, who adopted 9 mm firearms in 2015.^[20] In 2019, U.S. Customs and Border Protection adopted 9 mm Glock handguns, replacing the .40 S&W Heckler & Koch P2000 CBP was previously using.^[21]

The .40 S&W was originally loaded at subsonic velocity (984.25 ft/s (300.00 m/s)) with a 180 grains (11.7 g) bullet.^[17] Since its introduction, various loads have been created, with the majority being either 155, 165 or 180 gr (10.0, 10.7 or 11.7 g).^[22] However, there are some bullets with weights as light as 135 gr (8.7 g) and as heavy as 200 gr (13.0 g).^[23] Cor-Bon and Winchester both offer a 135 gr (8.7 g) JHP and Cor-Bon also offers a 140 gr (9.1 g) Barnes XPB hollow-point. Double Tap Ammo, based in Cedar City, Utah, loads a 135 gr (8.7 g) Nosler JHP, a 155 gr (10.0 g), 165 gr (10.7 g) and 180 gr (11.7 g) Speer Gold Dot hollow-point (marketed as "Bonded Defense"), a 180 gr (11.7 g) Hornady XTP JHP, and three different 200 gr (13.0 g) loads included a 200 gr (13 g) Full Metal Jacket (FMJ), a 200 gr (13 g) Hornady XTP JHP and Double Tap's own 200 gr (13 g) WFNGC (Wide Flat Nose Gas Check) hard cast lead bullet; the latter specifically designed for hunting and woods carry applications.

Case failure reports in older Glock pistols

The .40 S&W has been noted in a number of cartridge case failures involving older Glock pistols, due to the relatively large area of unsupported case head in those barrels, given its allowed chamber pressure level.^{[24][25]} The feed ramp on the Glock .40 S&W pistols is larger than on other Glocks, which leaves the rear bottom of the case unsupported, and it is in this unsupported area that the cases fail. Most, but not all, of the failures have occurred with reloaded or remanufactured ammunition.^[26] Cartridges loaded at or above the SAAMI pressure, or slightly oversized cases which fire slightly out of battery are often considered to be the cause of these failures,^[26] which are commonly referred to as "kaBooms" or "kB!" for short.^[26] While these case failures do not often injure the person holding the pistol, the venting of high-pressure gas tends to eject the magazine out of the magazine well in a spectacular fashion, and usually destroys the pistol. In some cases, the barrel will also fail, blowing the top of the chamber off.



Frontal view of .40 S&W cartridge, FMJ

Glock Inc. has since made chamber support modifications introduced in 2006 that has 99.9% eliminated this problem. Photos of barrels after 2006 show much more chamber support. The new Gen 5 models have increased the chamber support even further.



Rearview of .40 S&W pistol cartridge, FMJ

Synonyms

- .40 caliber
- .40 Auto
- 10×22 mm
- 10 mm Kurz (A moniker based on the .40 S&W being a "shorter version" of the 10 mm Auto, similar to how the .380 ACP (also known as 9 mm short or 9 mm Kurz) is shorter and lower powered but in caliber similar to the 9mm Parabellum)
- .40 Short & Weak (A moniker based on the cartridges overall under performance compared to 10 mm.)

See also

- 10 mm caliber
- 10 mm Auto
- .357 SIG
- 5.56×21mm PINDAD
- List of handgun cartridges
- Table of handgun and rifle cartridges
- List of firearms

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External links

- [Ballistics By The Inch .40 S&W Results \(http://www.ballisticsbytheinch.com/40sw.html\)](http://www.ballisticsbytheinch.com/40sw.html)
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